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Sourcing List — Automated Test Station

> 📄 **Print-ready PDF:** `docs/pdf/sourcing-list.pdf`

Shopping list for the breadboard prototype and first full build. Grouped by priority — Priority 1 blocks breadboarding and must be ordered before any bench work can start.

Legend: ✅ on hand · ⏳ ordered · 🛒 to source

Approximate prices are USD, mid-2026 retail. Use as ballparks, not quotes.

Priority 1 — Order Today (blocks breadboarding)

These parts are required before you can do anything on the breadboard. Order all of them in one cart.

#	Item	Qty	Status	Approx $	Notes
1	MAX31855 thermocouple amplifier breakout	3	🛒	~$5 ea	Amazon search: **"MAX31855 thermocouple amplifier breakout"** — Adafruit #269 or compatible clone. Exposes VCC, GND, DO, CS, CLK on a 5-pin SIP header. Verify breakout has 3.3V regulator or is marked 3.3V-only — the MAX31855 is NOT 5V.
2	Solderless breadboard, 830-point, full-size	1	🛒	~$8	Only if not already on hand. Amazon: **"830 point solderless breadboard"**. A half-size (400-point) will be too tight for 3 modules + protection resistors — get full-size.
3	E-stop mushroom button, NC, latching, 22mm panel mount	1	🛒	~$6	Amazon: **"22mm latching mushroom emergency stop button NC"** — red head, normally-closed contacts. Must be latching (twist-to-release), not momentary. This is a safety-critical component — don't use the cheapest listing; pick one with 10A/250VAC contact rating.
4	Green momentary pushbutton, NC, 22mm panel mount	1	🛒	~$3	Amazon: **"22mm momentary pushbutton green NC normally closed"** — manual START button wired to LOGO! I4. NC type allows fail-safe wiring.
5	Red momentary pushbutton, NC, 22mm panel mount	1	🛒	~$3	Amazon: **"22mm momentary pushbutton red NC normally closed"** — manual STOP button wired to LOGO! I5.

Priority 1 subtotal: ~$25–30

Priority 2 — Order Soon (needed for heater circuit)

Order these alongside Priority 1 or immediately after. The heater circuit cannot be tested or brought up without them.

#	Item	Qty	Status	Approx $	Notes
6	Fotek SSR-40DA solid-state relay	1	🛒	~$8–12	Amazon: **"SSR-40DA solid state relay"**. Spec: 3–32V DC control input, 24–480V AC load side, 40A, zero-crossing. At 150W/120VAC the load is 1.25A — well within rating. Must be genuine Fotek or a quality clone; cheap counterfeits have caused fires. Check seller reviews carefully.
7	150W 120VAC cartridge heater	1	🛒	~$12–18	Amazon: **"cartridge heater 150W 120V"** or search Omega part **"CIR-1012/120V"**. Common size: 5/16" (8mm) diameter × 3" (76mm) long. Verify 120VAC specifically — 240VAC units exist and look identical.
8	KSD301 bimetallic thermostat disc, 120°C, NC	2	🛒	~$1–2 ea	Amazon: **"KSD301 thermostat 120C NC normally closed"** — snap-disc type, mounts directly against heater body. 120°C trip point = hardware overtemp cutout wired to LOGO! I3. Buy 2: one installed, one spare.
9	Thermal cutoff fuse, 150°C, 10A, axial	5	🛒	~$1–2 ea	DigiKey/Mouser: search **"thermal cutoff fuse 150C 10A axial"**. Or Amazon: **"152C thermal fuse 10A"** — 152°C is a common standard rating close to 150°C. Spec: one-time non-resettable TCO type. Must be rated ≥10A. Clip directly to heater body — NOT in free air. Buy 5: 1 installed + 4 spares (a blown fuse means investigate; don't reuse).
10	Aluminium heatsink for SSR	1	🛒	~$5–8	Amazon: **"SSR heatsink aluminum solid state relay"** — look for a dedicated SSR heatsink with pre-drilled mounting holes matching 40A SSR footprint (57.5mm × 44mm typical). Without a heatsink the SSR overheats at sustained duty. A DIN-rail mounted extrusion with fins ≥50 cm² face area also works.

Priority 2 subtotal: ~$35–50

Priority 3 — Servo Vent Control

Order when you are ready to wire the vent servos. Not needed for initial heater bring-up.

#	Item	Qty	Status	Approx $	Notes
11	Tower Pro MG90S metal-gear servo (or SG90 plastic)	2	🛒	~$3–5 ea	Amazon: **"MG90S metal gear servo"** — metal gear recommended for repeated vent duty. SG90 plastic gear is fine for light loads. Standard 50Hz PWM, 500–2500µs pulse, 5V supply.
12	5V 2A BEC / DC-DC converter, 24V input	1	🛒	~$5	Amazon: **"24V to 5V DC DC converter 2A step down"** — powers servo rail from LOGO! Q2 24V relay output. Needs to handle 1A peak (2 servos at stall). Get a switching BEC, not a linear regulator (24→5V linear = 19V × 1A = 19W of heat = not good).

Priority 3 subtotal: ~$12–20

Already Have (verify before ordering)

Check your bench stock before adding these to a cart.

#	Item	Qty	Status	Notes
13	K-type thermocouples with leads	3	✅	One per MAX31855 channel — TC1 ambient, TC2 DUT, TC3 heater. Yellow wire = positive (ANSI standard).
14	Dupont jumper wires (M-F, F-F assortment)	—	✅	For breadboard-to-RPi header connections.
15	Siemens LOGO! 12/24RCE PLC	1	✅	Already installed / on hand.
16	24V DIN rail PSU (Meanwell HDR-60-24 or equivalent)	1	✅	Powers LOGO! and relay coils.
17	WeMos D1 Mini / NodeMCU ESP8266	1	✅	Used for status-display sketch. NodeMCU (27mm row-spacing) will work on a breadboard; D1 Mini preferred for clean mounting.
18	Arduino resistor kit — 100Ω and 10kΩ	—	✅	Probably — confirm you have at least 3× 10kΩ (CS pull-ups) and 3× 100Ω (protection resistors). If stock is low, order a 1/4W resistor assortment on Amazon: **"resistor assortment kit 1/4W 600 piece"**.
19	100nF (0.1µF) ceramic capacitors	3+	✅	Probably — confirm at least 3× 100nF for MAX31855 VCC decoupling.
20	SSD1306 128×64 OLED display (I²C)	1	✅	Check Arduino kit. If not on hand: Amazon **"SSD1306 0.96 inch OLED I2C"** ~$4.
21	Raspberry Pi 5 4GB	1	❓	**Required for GPIO.** Flask app runs on any host for dev, but SPI (MAX31855) and GPIO PWM (SSR, servos) need a Pi. See RPi note below.

Note on Raspberry Pi / SBC

You don't need an SBC at all for the prototype. The WeMos D1 Mini (ESP8266) already in your kit handles all the hardware I/O — SPI for MAX31855 thermocouples, PWM for the SSR heater and servo vents — and talks to the NUC over WiFi. The NUC runs the Flask app, Modbus TCP to the LOGO! PLC, USB to the HiPot, and TCP to the DC Load. Nothing to buy.

Set `"transport": "http"` in `plc/rig_config.json` (already done) and flash `esp8266/thermal-bridge/` to the D1 Mini.

If you want an SBC later: RPi 4B and 5 are available at MSRP ($55–60 for board only) through DigiKey, Mouser, and Arrow — all approved vendors. Check their sites directly; pricing should be at MSRP unlike the Amazon kit bundles. Used RPi units on Facebook Marketplace are also a practical option for a lab device — any Pi 4B or 5 works.

Note on Raspberry Pi

Use a Raspberry Pi 5 4GB. The Pi 4 is heavily inflated on Amazon ($125–150 for a kit). The Pi 5 is faster, more available, and at MSRP costs less.

Cheapest sourcing path (~$78 total):

Item	Source	Price
RPi 5 4GB board	PiShop.us or Adafruit ($60 MSRP)	$60
27W USB-C PD power supply	Any 27W+ USB-C PD adapter — check your desk first. Official RPi PSU from PiShop ~$12, or generic on Amazon ~$8	$8–12
32GB microSD A2-rated	Amazon: "Samsung EVO Select 32GB microSD"	$8
Case	None needed for prototype — bare board on bench is fine	$0

> If buying a kit, the **Vilros RPi 5 Complete Starter Kit** (vilros.com) is $134.99 on sale and includes board, 128GB SD, 27W PSU, aluminum case, and HDMI cables. Worth it if you want everything in one box — but $55 more than buying parts individually.

One extra setup step for Pi 5: the standard `RPi.GPIO` library does not support the Pi 5's GPIO chip. Install the drop-in replacement before running the Flask app:

```
pip install rpi-lgpio --break-system-packages
```

This provides a fully compatible `RPi.GPIO` API on the Pi 5. No code changes required — just install it before `requirements.txt`.

SPI must be enabled: `sudo raspi-config → Interface Options → SPI → Enable → reboot`.

Total Cost Estimate

Priority bucket	Subtotal
Priority 1 — breadboard unblock	~$25–30
Priority 2 — heater circuit	~$35–50
Priority 3 — servos	~$12–20
RPi 5 4GB + PSU + SD (if needed)	~$78
Total to source	**~$150–178**

Minimum spend to start breadboarding (Priority 1 only, RPi on hand): **~$25–30**.
